Applications of Statistical Modeling: Module #1, Lecture #2

Software for Fitting Multilevel Models

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Lecture Overview

- Today, you will see an overview of popular general-purpose statistical software alternatives for fitting multilevel models
- We will consider software for both linear and generalized linear mixed-effects models
- Conceptual material from Lecture #1 is important for understanding the various software options and alternatives; we'll keep both sets of slides open simultaneously!

Lecture Overview, cont'd

- This lecture will introduce the syntax for fitting these models in various software packages
- Next week, we will apply these procedures to a real data set in an investigation of **interviewer effects**, and explore the output generated by the procedures
- **You should start practicing with the use of these procedures, preferably with a data set in mind for the final project**

R Software

- The most readily available general-purpose statistical software (because it's free)
- Lots of good tutorials available and new menu-driven interfaces (e.g., RStudio)
- Primary multilevel modeling functions:
  - `lmer()` and `glmer()` (in the **`lme4`** package)
  - `lme()` (in the **`nlme`** package; part of base install)
  - Others [e.g., `hglm()`]

R Software, cont'd

- The `lme()` function is good for fitting models to normal outcomes, and offers more options
- The `glmer()` and `lmer()` functions are good for fitting models to non-normal outcomes and models with ***crossed*** random effects; but there aren't as many options (and possibly no p-values…)
- We will focus on the `g/lmer()` functions because they handle a more general set of models; some testing is made possible by also downloading the contributed **`lmerTest`** package

R: Required Data Structure

- The procedures in R generally require a long or "vertical" data structure, with multiple rows per cluster in the data set, and ID codes
- Example (could be Cluster / ID instead):

Subject	Time	DV	IV1	IV2	Male	
1	1		15.2	1	1	1
1	2		12.3	0	1	1
1	3		12.2	1	0	1
2	1		16.3	1	2	0 ...

R: Required Data Structure

- A given data frame object therefore needs to have a "vertical" data structure
- Easiest to import data sets from other packages using functions in the **`tidyverse/foreign/haven`** packages, or import delimited text files
- The `reshape` function changes data structure (e.g., wide to long)
- The following syntax assumes that a data frame object with the vertical structure (`datan`) has been created and is available

lmer() syntax

```
> library(lmerTest)
> example.model <- lmer(vsae ~ iv1 + time + factor(male)
+ (time | subject),
REML = T,
data = datan)
> summary(example.model)
> ranef(example.model)
> ranova(example.model)
```

Load the lmerTest package, enabling approximate tests of fixed-effect parameters

Coefficient for TIME and intercept (by default) randomly vary across subjects, with UN structure

First part of model formula: DV, and fixed effects

Estimation method (REML), and data frame object

Use the summary() function to view estimates, tests, and information criteria

Generate multi-parameter tests for fixed effects (also used for likelihood ratio tests, where two model fit objects are input)

Print EBLUPs for all random coefficients

Other `lmer()` options

- Maximum Likelihood Estimation (e.g., for testing fixed effects):

```
REML = F
```

- Mixture-based Likelihood Ratio Tests for variance components in intercept-only models (requires `lmerTest`):

```
rand(example.model)
```

- Heterogeneous Residual Covariance Parameters: use `lme()`!

```
library(nlme)
examp2 <- lme(dv ~ iv1 + time + factor(male),
      random = ~time | subject, datan, method =
"REML", weights = varIdent(form = ~1 | male))
```

Other `lmer()` options

- Pairwise comparisons of estimated marginal means for all levels of all factors included in the model, after a model has been fitted (needs `lmerTest`):

```
> difflsmeans(example.model)
```

- Model diagnostics:
  - Use `resid()` and `fitted()` functions to create new variables, and then apply various plotting functions [e.g., `plot()`] to them
  - Influence diagnostics take some work; available as part of `nlmeU` package

Logit model: `glmer()` syntax

```
> logit.model <- glmer(bindv ~ iv1 + time + factor(male)
+ (1 | subject),
family = binomial,
nAGQ = 10,
data = datan)
> print(logit.model)
```

Specify that the intercept (1) randomly varies across subjects

First part of model formula: binary DV, and fixed effects

Distribution of DV (binomial implies logit link function)

Number of integration points for Adaptive Gaussian Quadrature (approximating the likelihood function)

Use the print() function to view estimates, tests, and information criteria

For an excellent fully-worked example with visualization, see:

https://stats.idre.ucla.edu/r/dae/mixed-effects-logistic-regression/

More Details on R Procedures

- Learning the syntax is not that hard; important to consult reference manuals
- R makes many powerful graphical procedures available (see the prior worked example…)
- Some error messages can be a bit ambiguous
- The `nAGQ` argument defaults to 1, which is Laplace Approximation; higher integers = more accurate approximation = slower estimation times
- **You can generally do everything that SAS, SPSS, and Stata can do, for free; this just takes practice!**

Stata Software

- Readily available at MPSDS and JPSM
- Primary multilevel modeling procedures:
  - `mixed` (for normal outcomes)
  - `melogit, meologit, etc.` (for non-normal outcomes)
- **Advantages:** highly-integrated general-purpose package with powerful data management and graphical features; easy-to-use, point-and-click interface; very user-friendly and fast

Stata: Required Data Structure

- Same: Needs a “vertical” data structure
- If needed, Stata has a built-in data restructuring command (`reshape`) that is also among the best of the major statistical packages
- The following syntax assumes that a Stata data file having the vertical structure is open

mixed syntax

```
. mixed dv iv1 time i.iv2 i.male
|| subject: time,
covariance(unstruct)
variance
reml
```

First part of model formula: DV, and fixed effects (note the use of i. for categorical IVs)

Second level (||): Coefficients for time and intercept (by default) randomly vary across subjects

Use unstructured variance-covariance matrix for random effects

Print estimated **variance** components, not standard deviations (default)

Estimation method (REML)

```
. estat ic
. test 1.iv2 2.iv2
. predict eblups*, reffects
```

Display information criteria

Generate multi-parameter Wald tests for fixed effects (the numbers are values of categories, corresponding to estimated fixed effects)

Save predicted values of random effects in new variables

Other `mixed` options

- For maximum likelihood estimation, just leave out the `reml` argument (ML is the default)
- Heterogeneous Residual Covariance Parameters:

```
. mixed dv iv1 time i.male || subject: time,
covariance(unstruct) variance reml
residuals(independent, by(male))
```

- Post-hoc pairwise comparisons of means:

```
. margins male, pwcompare(effects)
```

- After `margins`, use `marginsplot` for graphing!

Other `mixed` options

- Model Diagnostics:
  - Accomplished using post-estimation and plotting commands, after fitting a model using `mixed`
  - `predict st_resid, rstandard`
  - `predict predvals, fitted`
  - `twoway (scatter st_resid predvals), yline(0)`
  - `qnorm st_resid`
  - `qnorm eblups1`
  - `qnorm eblups2`

Logit model: `melogit` syntax

```
. melogit bindv i.iv2 time i.male
|| subject: ,
covariance(identity)
intmethod(mvaghermite)
intpoints(7)
```

First part of model formula: DV, and fixed effects (note the use of i. for categorical IVs)

Second level (||): Only intercept is allowed to randomly vary across subjects (no variables after the colon)

Only one random effect; no need for unstructured

Approximation method (AGQ) and number of integration points; both are defaults

```
. estat ic
. test 1.iv2 2.iv2
. predict eblups*, reffects
```

Display information criteria

Generate multi-parameter Wald tests for fixed effects (the numbers are values of categories, corresponding to estimated fixed effects)

Save predicted values of random effects in new variables

SAS Software

- Readily available at MPSDS and JPSM, and in the Federal Statistical System
- Primary multilevel modeling procedures:
  - `proc mixed` (for normal outcomes)
  - `proc glimmix` (for non-normal outcomes)
  - `proc nlmixed` (for non-linear models)
- **Advantages:** integrated general-purpose package with powerful data management features; among most advanced procedures

SAS: Required Data Structure

- Same data structure required as in R:
  - Multiple rows per cluster (or subject)
  - Cluster / Subject ID variable
  - See Slide 6 for an example
- The next few slides assume that a SAS data set named `datan` has been created and saved in the `libn` library

proc mixed syntax

```
proc mixed data = libn.datan covtest;

    class subject male cattime;

    model dv = time iv1 male / solution;

    random int time / subject = subject
        type = un g gcorr v vcorr solution;

    repeated cattime / subject = subject
        type = ar(1);

run;
```

SAS data set, and SE for estimated covariance parameters

Categorical variables, including the subject ID

Dependent variable, and fixed effects to estimate

Which coefficients should be allowed to randomly vary across subjects (int, time)? What is structure of *D* matrix (type = un)? Also display estimated *D* and *V* matrices, and EBLUPs

For longitudinal data, indicate categorical version of time variable (cattime), subject ID, and type of *R* matrix (e.g., AR(1))

Other `proc mixed` options

- Maximum Likelihood Estimation (e.g., for testing fixed effects):

```
proc mixed data = libn.datan covtest method = ml;
```

- Heterogeneous Residual Covariance Parameters (also possible for `random`):

```
repeated cattime / subject = subject
     type = ar(1) group = male;
```

- Post-hoc Pairwise Comparisons of Means:

```
lsmeans male / adjust = tukey;
```

Other `proc mixed` options

- Model Diagnostics:

```
ods graphics on;
ods rtf file = "c:\temp\diags.rtf" style =
   journal;
ods exclude influence;
proc mixed data = libn.datan plots =
  (residualpanel boxplot influencestatpanel);
  class subject male cattime;
  model dv = time iv1 male / solution residual
    outpred = pdat1 influence (iter = 5 effect =
      subject est);
ods output solutionR = eblupsdat ;
```

proc glimmix syntax

```
proc glimmix data = libn.datan;

    class subject male;

    model bindv (event = "1") = time iv1 male /
        dist = binary link = logit
        solution cl;

    random int / subject = subject solution;

    covtest glm;

run;
```

SAS data set

Categorical variables, including the subject ID

Binary dependent variable, fixed effects to estimate, distribution of DV, and link function (logit model)

Which coefficients should be allowed to randomly vary across subjects (int)? Also compute EBLUPs

Appropriate likelihood ratio tests for variance components based on mixtures of chi-square distributions

More details on SAS procedures

- SAS Online Help (Best Resource!)
- West, Welch, and Galecki (2022)
- *SAS for Mixed Models* (Littell et al.)
- Several alternative estimation procedures are available for `proc glimmix` (pseudo-likelihood, ML with Laplace Approximation, ML with Adaptive Quadrature); more on this choice later in lecture

So What Estimation Method Should Be Used for GLMMs?

- As shown in the previous slides, there are many possible methods for estimating the parameters in GLMMs
- When using Adaptive Gaussian Quadrature (AGQ) to approximate the likelihood (probably the most popular method), more integration points means better approximation, but also more computation time

So What Estimation Method Should Be Used for GLMMs?

- Kim et al. (2013, *TAS;* see Canvas) suggest that Laplace approximation (# integration points = 1) works well, but AGQ with more points will be better for smaller sample sizes (e.g., longitudinal data)
- Laplace is far superior to AGQ when working with large sample sizes, but can be biased
- Pseudo-likelihood / penalized quasi-likelihood estimation can lead to considerable bias!

Other Software Options

- MLwiN, HLM (multilevel specification)
- SuperMix, aML
- ASReml, SAS `proc hpmixed` (for large data sets)
- Mostly boutique packages that do not also offer data management capabilities
- Stata, `glimmix`, HLM, and MLwiN currently offer correct theoretical options for **weighted estimation** of multilevel models: more on this in two weeks
- For R users working with weights: the relatively new **`svylme`** package

Bayesian Estimation

- Stan: https://mc-stan.org/users/interfaces/
- R packages: RStanArm; brms (see https://websites.umich.edu/~bwest/brms_example.txt for an example)
- A new Multilevel Regression and Poststratification (MRP) interface: https://mrpinterface.shinyapps.io/shinymrp/

Thinking Ahead…

- Now that we have discussed conceptual background and software for fitting multilevel models, we will consider a case study next week, focusing on estimation of interviewer variance in the ESS Data
- Please continue practicing with these procedures using your own data! We will have time for more questions next week.